Maxmilian Duarte

Python Backend Engineer

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PROFESSIONAL SUMMARY

Python backend engineer with 4 years of production experience building REST APIs, browser automation that runs unattended, and data pipelines on advanced SQL. I own systems end to end: design, implementation, tests, and deployment with Docker on Linux.

SKILLS

Backend: Python, Flask, FastAPI, REST APIs, asyncio / aiohttp, OAuth2 / PKCE, Server-Sent Events
Data: MySQL, SQLite, MongoDB, advanced SQL, stored procedures, ETL
Infra: Docker / Compose, Linux, systemd, nginx, Proxmox / LXC, GitHub Actions, Uptime Kuma, scheduled backups
Quality: pytest, end-to-end tests, type hints, production support (L3), log analysis
Also: TypeScript / Next.js, Vue 3, Kotlin / Jetpack Compose, Selenium / Playwright, FFmpeg

WORK EXPERIENCE

Independent Developer — Jul 2026 – Present
Own products and open source — Remote

- Building MaxTools Radio and cnabkit, an open-source Python library for Brazilian bank files (CNAB 240/400, OFX), on self-managed infrastructure: Proxmox, Docker, nginx, Uptime Kuma, and weekly backups.

Software Engineer (Python, Backend and Data) — Sep 2024 – May 2026
Netbiis | Data Driven Technologies — Remote

- Built and owned an internal Flask and SQL web app for about 32 daily campaign jobs, with automated QA fully covering 10 to 15.
- Cut each campaign job from about 20 to 12 minutes (−40%) with the app's support tools, used on all 32.
- Wrote and optimized complex SQL and stored procedures for high-volume extraction and segmentation in Adobe Campaign.
- Provided L3 support for production automation pipelines: log-based root-cause analysis, fixes, and follow-up.

Backend Developer, Independent Consultant — Feb 2024 – Aug 2024
HG7 Group (Insurance Brokerages and a Clinic) — Hybrid

- Built an OFX bank-statement import and reconciliation system (Python, Flask, MySQL) across three iterations, embedded in the company's PHP ERP and feeding its income statement, replacing line-by-line spreadsheet reconciliation.
- Built async ETL from RD Station CRM into MySQL for Power BI (10 parallel requests per batch, paced to the 120 requests/min rate limit) and from Exact Spotter into MongoDB.
- Built a bot that reads answered calls from the 3CX phone system and creates RD Station leads assigned to the right salesperson.

Python Developer (Internship) — Aug 2022 – Feb 2024
Plug Benefícios — São Paulo

- Built and maintained 15 production bots (Python, Selenium) processing hundreds of beneficiary changes a month across 4 health-insurer portals from a MySQL ticket queue, plus invoices, slips, and payslips.
- For ExportSul, a client brokerage, built bots on the Akad, Sompo, and Tokio Marine portals that register export shipments and download invoices, with per-record status in MySQL so runs resume where they stopped.
- Packaged shared code (browser setup, audio-based reCAPTCHA solving, FTP upload) into an internal Python wheel, later reused across other client projects.

Tech Instructor — Oct 2021 – Feb 2022
Microcamp Tecnologia — São Paulo

- Taught programming logic, algorithms, and Python to 50+ students.

PROJECTS

Case studies at maxmilian02.github.io

MaxTools Radio: Python bot streaming 48 kHz PCM through a real-time DSP chain, Flask API with SSE and OAuth accounts (Google, Discord, Spotify), web app, and Kotlin / Jetpack Compose Android app. 1,264 Python tests and 472 Android unit tests.

MaxTools: FastAPI plugin platform where a registry auto-discovers tool modules and a Next.js dashboard renders from their manifest. Async jobs with progress, API-key auth, TTL cleanup, and frame-accurate video trimming over HTTP Range.

LANGUAGES

Portuguese (native), English (advanced: reading, writing, technical documentation)

EDUCATION

B.Sc. Computer Engineering, Impacta Tecnologia (expected 2027)